SECP2753-01: DATA MINING

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ASSIGNMENT 1

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1.0 Introduction

Data mining is a fundamental process in discovering patterns and useful insights from large datasets. The quality of results derived from data mining depends significantly on the understanding and preparation of the data. Hence, data exploration and preprocessing are essential steps before building any predictive model.

This report aims to conduct a detailed exploration of two selected datasets—Abalone and Mushroom—to identify key patterns and insights. Additionally, the report focuses on the preprocessing techniques of normalization and discretization, which are crucial to ensure the data is well-suited for subsequent analysis. The Abalone dataset will be used to illustrate these preprocessing steps.

2.0 Dataset Exploration

<u>Mushroom Dataset</u>

The first dataset we chose is the Mushroom dataset. It is a multivariate dataset that contains a total of 8124 instances and includes 22 categorical attributes that describe the physical characteristics of mushrooms such as cap-shape, cap-surface, cap-colour and others. All attributes are nominal(categorical) types and each one is represented with a single character code making it suitable for classification tasks. The primary goal of the dataset is to classify mushrooms as either edible(e) or poisonous(p) based on these features. Based on the UCI Machine Learning Repository, there are missing values for the "stalk-root" attribute.

There are some key insights from this dataset. First and foremost, odor is a strong indicator of poisonous mushrooms. Odor is one of the most predictive features for determining edibility. Other than that, simple physical traits like bruising help in safe identification also. Mushrooms that bruise easily are more likely to be edible. Next, gill characteristics also help in classification. Mushrooms with broad gills and with white or pink gill colour lean toward edible while mushrooms with narrow gills with dark or black colours tend to be poisonous.

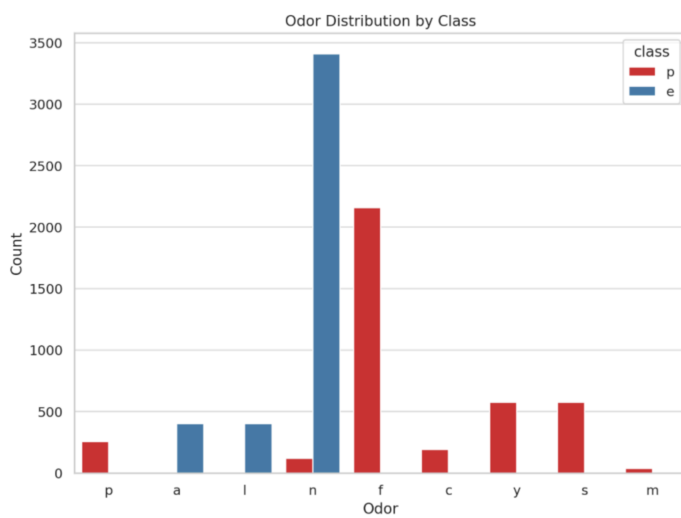
Key Insights:

1. Class Distribution

   The mushroom dataset has relatively balanced distribution between edible and poisonous mushrooms:
   Edible mushrooms: approximately 51.8% (4208 samples)
   Poisonous mushrooms: approximately 48.2% (3916 samples)
   This balanced distribution is beneficial for classification tasks as it prevents model bias toward the majority class.

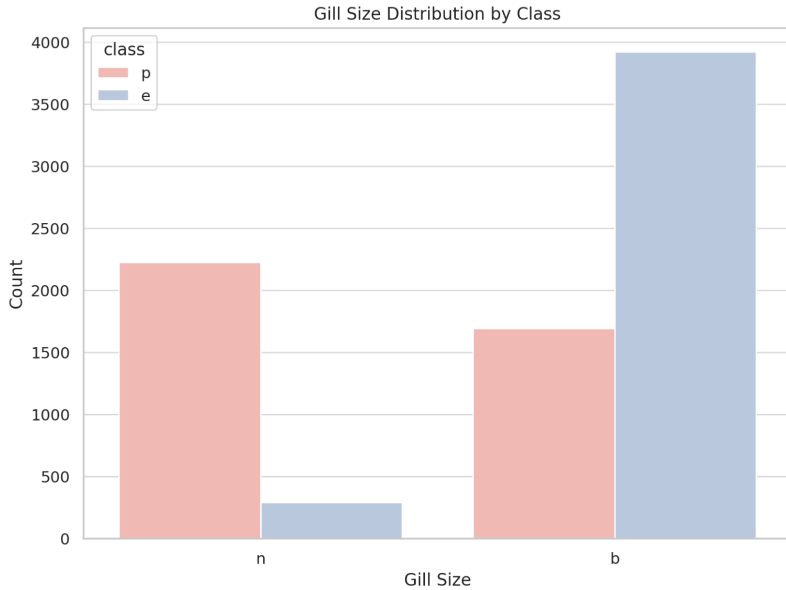
2. Odor as a Strong Predictor

One of the most significant insights from the dataset is that odor is an exceptionally strong predictor of mushroom edibility. Mushrooms with odors described as almond (a) and anise (l) are consistently edible while mushrooms with odors described as creosote (c), fishy (y), foul (f), musty (m), pungent (p), and spicy (s) are consistently poisonous. However, mushrooms with no odor (n) can be either edible or poisonous.



3. Gill size

Gill size proves to be a meaningful attribute in determining mushroom edibility. In the dataset, gill size is categorized as either broad (b) or narrow (n). A significant majority of edible mushrooms possess broad gills, while narrow gills are far more common among poisonous mushrooms. This difference provides a useful insight in mushroom identification to encounter a mushroom with broad gills may indicate a higher likelihood of being edible, whereas narrow gills should raise caution. However, it is important to note that gill size alone should not be used as a definitive predictor without considering other attributes, as exceptions do exist.



Abalone Dataset

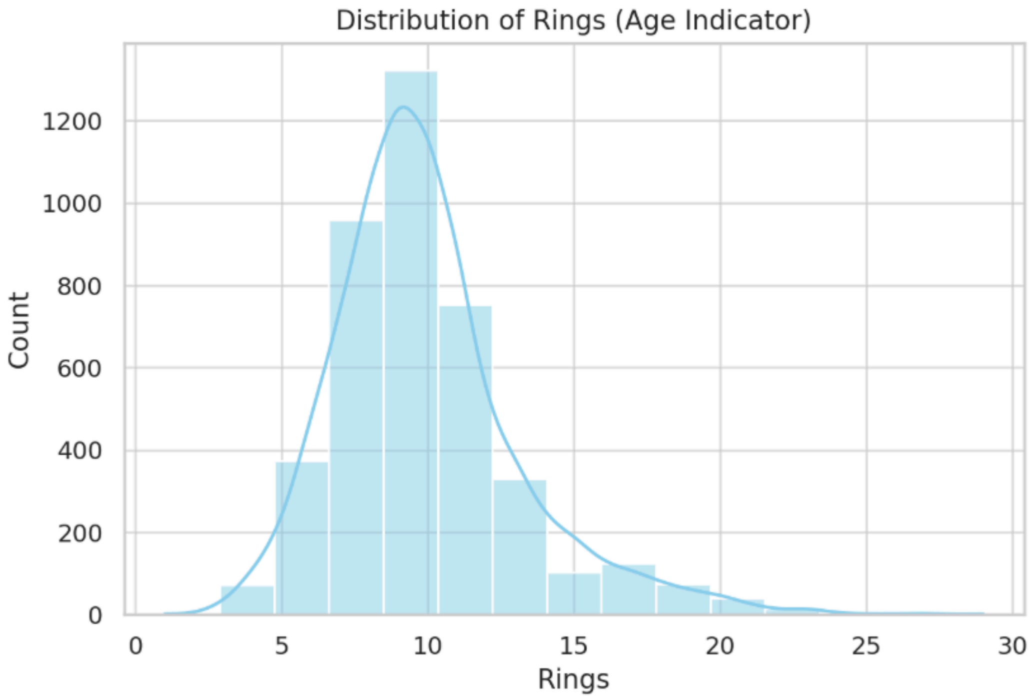
The second dataset we choose is the Abalone dataset that sourced from the UCI Machine Learning Repository also. It is designed to support the prediction of abalone age using physical measurements. Age is approximated by the formula: Age = Rings + 1.5.

The Abalone dataset consists of 4,177 records, each describing a single abalone, and includes 9 attributes. These features include both categorical and numerical variables that collectively detail the biological structure of the abalone. The most important feature for this task is "Rings," which serves as the target variable for age estimation. Other features describe physical dimensions and weights, such as Length, Diameter, Height, Whole Weight, Shucked Weight, Viscera Weight, and Shell Weight. Additionally, the "Sex" attribute distinguishes each abalone as male (M), female (F), or infant (I).

Key Insights:

1. Distribution of Rings

   The exploratory analysis of the Abalone dataset highlights several key patterns. One of the key attributes in the dataset is **Rings**, which is used to estimate the age of an abalone. When examining the distribution of the Rings variable, it becomes clear that the majority of abalones fall within the range of 8 to 12 rings. This indicates that most of the specimens in the dataset are middle-aged, roughly between 9.5 and 13.5 years old, using the formula **age = Rings+ 1.5**. There are relatively fewer very young or very old abalones in the data, which could make it challenging for a model to learn to predict these rare age groups accurately.



2. Physical Differences by Sex

The Sex attribute (Male, Female, Infant) shows clear physical differences. Infants are the smallest and lightest, as expected. Females tend to be the largest and heaviest, followed closely by males. These differences are especially evident in features like Length and Whole Weight, making Sex a valuable predictor in modeling. Length was analyzed further due to its strong correlation with both age and size. On average, females have the greatest length, followed by males, with infants significantly smaller. This is illustrated in **Figure 1** (Boxplot of Length by Sex).

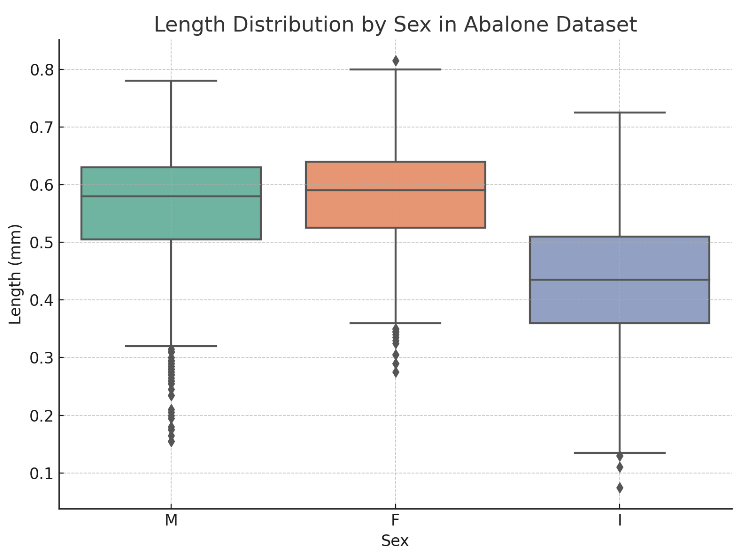


Figure 1: Boxplot of Length by Sex

This plot visually demonstrates differences in shell length across the Sex categories, illustrating that females have the highest average length followed by males, with infants showing the lowest values.

3. Whole Weight by Sex

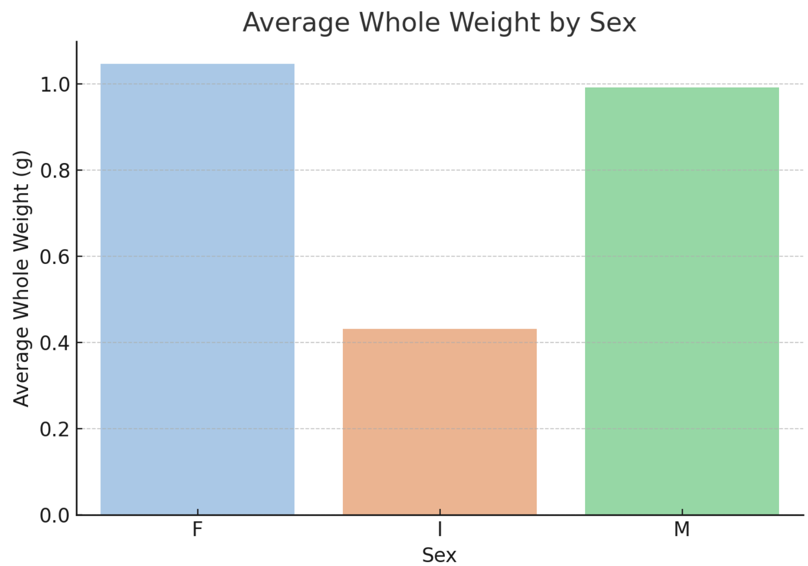


Figure 2: Bar Plot of Average Whole Weight by Sex

A similar pattern is observed when analyzing Whole Weight across the different Sex categories. As shown in **Figure 2**, females have the highest average whole weight, reinforcing their dominance in both size and mass. Males show a slightly lower average weight, while infants again fall significantly below both groups. This further supports the conclusion that the Sex attribute is a critical factor when modeling the physical maturity of abalones. Including is as a feature is likely to enhance the performance of predictive models that aim to estimate age or classify specimens based on biological structure.

3.0 Preprocessing Dataset : Abalone

The Abalone dataset contains several physical measurements of abalone, such as length, diameter, height, whole weight, shucked weight, viscera weight, shell weight, and the number of rings (which is used to estimate the age). The objective is often to predict the age of an abalone using its physical features. To prepare this dataset for data mining models, we apply key preprocessing techniques: **normalization** and **discretization**.

3.1 Normalization Techniques

Normalization is a technique to rescale numerical data values into a standard range—commonly [0, 1]. This ensures that features with larger magnitudes do not outperform those with smaller ones, which is especially important in algorithms like k-Nearest Neighbors and Support Vector Machine.

In the Abalone dataset:

- **Length** ranges from 0.075 to 0.815
- **Whole weight** ranges from 0.002 to 2.825

In this report, we use Min-Max Normalization, which maps an original value, v to a normalised value, v' using the formula:

$$Normalised\ Value, v' = \frac{v - min_A}{max_A - min_A}(new\_max_A - new\_min_A) + new\_min_A$$

Where:

- v is the original value
- min_A, max_A are the original min and max of feature A
- new_min_A, new_max_A are the desired scaling bounds

Example: Normalize Length = 0.5 to [0, 1]

Given:

- min_A = 0.075, max_A = 0.815
- new_min_A = 0, new_max_A = 1

$$v' = \frac{0.5 - 0.075}{0.815 - 0.075} \times (1 - 0) + 0 = \frac{0.425}{0.74} \approx 0.574$$

After normalization, the Length value of 0.5 becomes approximately 0.574.

3.2 Discretization Techniques

Discretization converts continuous numerical values into categorical bins or intervals. This can improve the interpretability of models and it is useful for algorithms that perform better with categorical inputs.

The **Rings** attribute (used to estimate age) is a continuous integer value. Instead of predicting exact ring counts, we can simplify the problem by grouping them into age categories.

Example: Discretizing using Equal-Width Binning

Step 1: Identify the Range

We'll use the **"Rings"** attribute from the Abalone dataset.

- **Lowest Value (A)** = 1
- **Highest Value (B)** = 29
- **Number of bins (N)** = 3

Step 2: Calculate Bin Width

Bin Width = $\frac{(B-A)}{N}$

$= \frac{(29-1)}{3}$

$= \frac{28}{3}$

$= 9.33 \approx 9$

Step 3: Define Bin Ranges

Now we use the bin width to define the intervals:

1. **Bin 1 (Young)**: $1 \leq \text{Rings} < 10$
2. **Bin 2 (Middle-aged)**: $10 \leq \text{Rings} < 19$
3. **Bin 3 (Old)**: $19 \leq \text{Rings} \leq 29$

Step 4: Assign a Value to a Bin

Let's say we have a value: **Rings = 15**
This falls in **Bin 2**, so we assign it to: Age Group = Middle-aged

4.0 Discussion

In the first part of this assignment, we did research and explored significant insights on both mushroom and abalone datasets. For the Mushroom dataset, it became clear that attributes such as odor and gill size are highly predictive of edibility. Odor, in particular, showed a strong and consistent relationship with mushroom toxicity, making it a vital feature for classification tasks. Additionally, physical features like bruises and gill characteristics offered valuable indicators for edibility, showcasing how a combination of attributes can support more reliable decision making.

However for the Abalone dataset, it revealed meaningful patterns between the physical measurements and the age of abalones. The distribution of the Rings attribute indicated that most abalones are middle-aged, which could influence model performance by causing bias toward predicting middle-aged specimens. Furthermore, the analysis of the Sex attribute demonstrated clear differences in size and weight among male, female, and infant abalones.

Furthermore, preprocessing was essential in preparing the Abalone dataset for analysis. Normalization helped scale features like Length and Whole Weight to a uniform range, preventing larger values from dominating algorithms that rely on distance calculations. This ensured fair contribution from all numerical attributes, such as Height and Diameter. For instance, without normalization, the model might give highlight to the Whole Weight simply because of its larger numerical range compared to Height or Diameter.

Discretization was applied to the Rings attribute after that, grouping it into Young, Middle-aged, and Old categories using equal-width binning. This matched the observed distribution, where most abalones had 8–12 rings, and simplified age prediction into a classification task. These steps improved data quality and supported insights from the exploration, such as age trends and sex-based physical differences to make the analytics clearly.

5.0 Conclusion

This report explored and analyzed two datasets, Mushroom and Abalone, to understand their attributes and the impact of preprocessing techniques. The Mushroom dataset, composed entirely of categorical features, was ideal for classification tasks. By examining key attributes like odor and gill characteristics, we identified strong predictors of edibility, demonstrating how categorical data can be leveraged for classification models.

In contrast, the Abalone dataset required preprocessing due to its numerical attributes. Normalization was applied to rescale features such as Length and Whole Weight, ensuring fair contribution across different magnitudes, which is essential for distance-based algorithms like k-NN. Discretization transformed the Rings attribute into categorical age groups, simplifying the problem from regression to classification and improving interpretability. These preprocessing steps are crucial for enhancing data quality and ensuring better model performance.

Through this analysis, we highlighted the importance of data exploration and preprocessing in preparing datasets for predictive modeling. Future work could involve applying machine learning techniques to classify mushrooms or predict abalone age more accurately and precisely while experimenting with alternative preprocessing methods to refine results.

6.0 Reference

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